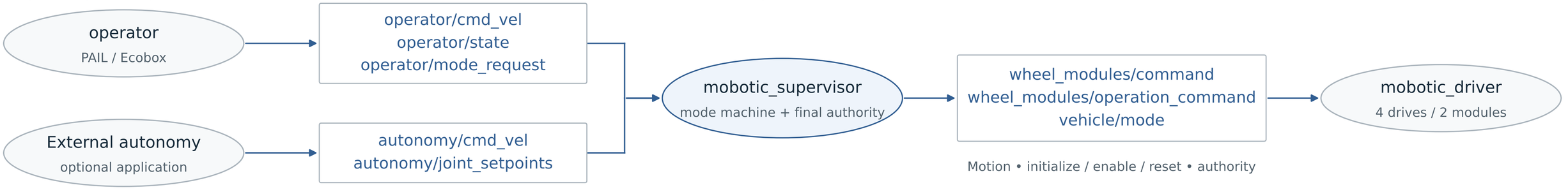
MoboTerra | ROS 2 node & topic graph

Two wheel modules: front_right + rear_left | PAIL / Ecobox over CAN | Namespace: /moboterra

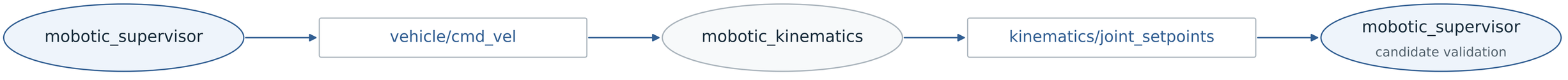
Source-defined hardware profile • repeated nodes are the same ROS node • not a live rqt capture

01 Command flow

Only the supervisor publishes the final motion array. PAIL has no hold-to-drive button; neutral rearm is required.



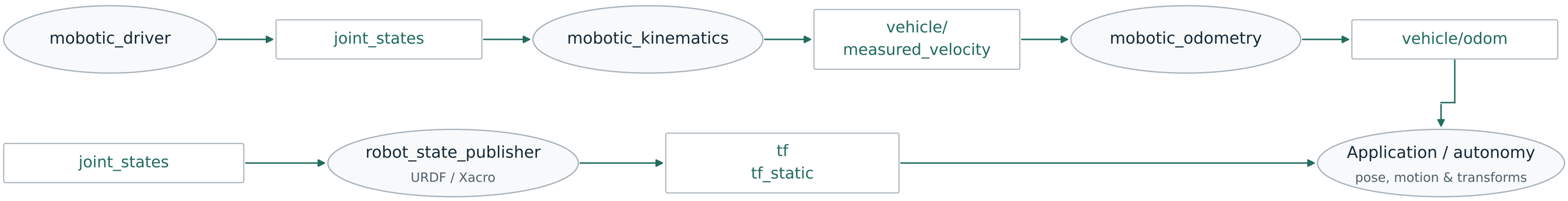
Velocity conversion loop



MANUAL / AUTONOMOUS use this loop. KINEMATICOFF validates autonomy joint candidates directly in the supervisor.

02 Measured feedback & transforms

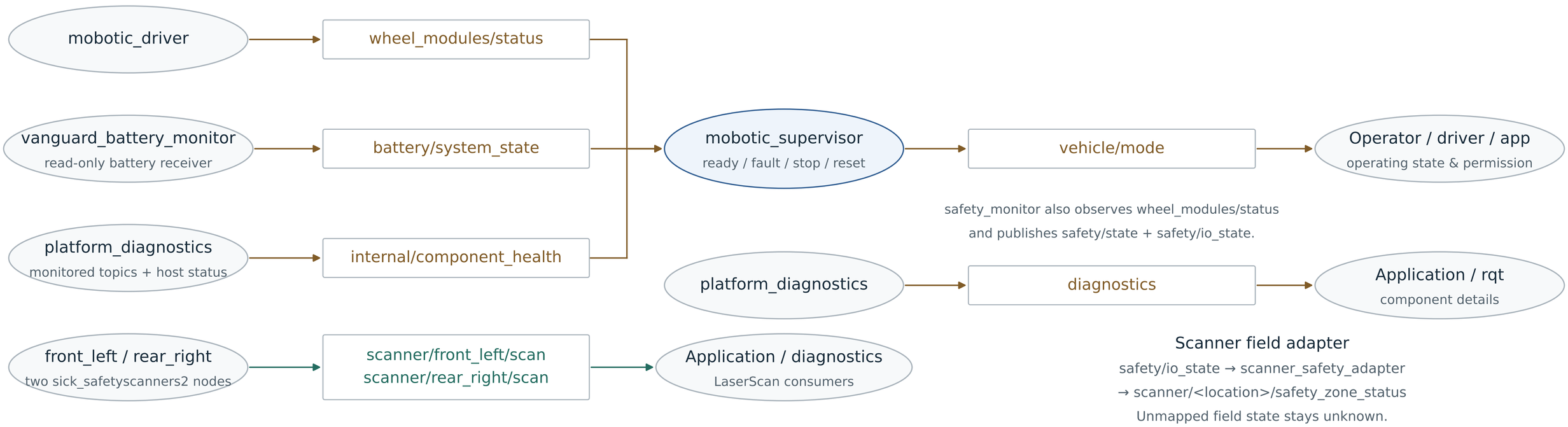
Position and velocity become valid after supervised drive enable. Feedback stays separate from command candidates.



joint_states and vehicle/measured_velocity also feed the supervisor for feedback freshness and measured standstill.

03 Readiness, health & scanner data

Fresh battery, operator and drive status gate enable. Complete feedback and required component health gate motion.



04 Physical connections

The two-module profile uses hardwired Siemens STO; there is no FlexiSoft TCP gateway in this configuration.

DRIVE + OPERATOR BUS

can0 · 250 kbit/s · ros2_socketcan

can_rx / can_tx ↔ driver + operator

Elmo nodes 3 / 4 / 7 / 8 · PAIL 10 · master 11

BATTERY BUS

can1 · 500 kbit/s · receive only

battery_can/can_rx → battery monitor

battery/state + battery/system_state

Independent physical chain: scanners → Siemens safety controller → drive STO

Ellipses = ROS nodes / external consumers | Rectangles = topics (grouped where shown) | Services and /rosout omitted